

Responsive Web Design

Assignment 3

Media Queries + Bootstrap Grid

OFFLINE

One page for a fictional campus club where students make things together and take a break from screens.

Student	Group
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Report date: 3 October 2026

Format: individual assignment; one responsive page

Source and published page

GitHub repository

<https://github.com/xnionn/WEB-midterm>

Assignment source folder

<https://github.com/xnionn/WEB-midterm/tree/main/assignment3>

Published website

<https://xnionn.github.io/WEB-midterm/assignment3/>

The assignment is stored separately in **assignment3/**. The OFFLINE concept, local fonts and visual language are adapted from the existing midterm project; this page has its own markup, stylesheet and responsive implementation.

One grid, three arrangements

The repeated ritual cards use native CSS Grid. The section contains only project-specific classes: there are no Bootstrap grid or utility classes inside it. Its fluid container grows with available space and the cards use fractional tracks instead of fixed pixel widths.

Viewport	Custom rule	Result
375 px	Base CSS	One card per row
768 px	@media (min-width: 48rem)	Two cards per row
1280 px	@media (min-width: 64rem)	Three cards per row

How mobile-first works

The base rule is the phone layout. A min-width query adds an override once the viewport is wide enough. At 48rem, the heading grows from 2rem to 2.5rem, section padding from 2.75rem to 3.5rem, and grid gap from 1.5rem to 2rem. The query changes both the layout and its spacing.

```
.ritual-grid {  
  display: grid;  
  grid-template-columns: 1fr;  
}  
@media (min-width: 48rem) {  
  .ritual-grid { grid-template-columns: repeat(2, minmax(0, 1fr)); }  
}  
@media (min-width: 64rem) {  
  .ritual-grid { grid-template-columns: repeat(3, minmax(0, 1fr)); }  
}
```

The excerpt isolates the layout rules; style.css also contains the section spacing, typography, colors and interactive states. There are no floats or absolutely positioned layout elements in the section.

Phone and tablet

Left: 375 px, cropped to the three stacked items. Right: 768 px, the full section with two columns. The same cards stay in document order.

01 / LOOK

Draw a bad chair.

Pick something nearby. Draw it without erasing.
The point is to notice, not to impress.

A pencil + any scrap of paper

02 / LISTEN

Ask a better question.

"What made you laugh this week?" Put both phones away. Let the answer take its time.

One friend + your full attention

03 / WANDER

Take the long way.

Walk a familiar route. Find three things you have never noticed. No photos required.

Comfortable shoes + curiosity

01 / START SMALL

Ten minutes. A different kind of break.

No booking, no special kit. Borrow one of these for the space between your classes.

01 / LOOK

Draw a bad chair.

Pick something nearby. Draw it without erasing.
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"What made you laugh this week?" Put both phones away. Let the answer take its time.

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Laptop: the three-column grid

1280 px. The custom 64rem query creates three tracks, preserving the section's own typography and spacing.

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A shared grid for the rest

The other page sections use Bootstrap 5.3.8 from the official CDN: a container limits the line length, a row groups columns, and responsive col-* classes control their widths. The local stylesheet is linked after Bootstrap so the club's colors, buttons and card details can override the defaults.

Requirement	Implementation on this page
Responsive cards	col-12 col-md-6 col-lg-4: one card below 768 px, two from 768 px, three from 992 px.
Stacked columns	The hero uses responsive columns and order-* classes. Essential introduction text comes before the illustration on a phone.
Visibility	The hero note uses d-none d-md-block, hiding it below 768 px. The navbar location uses d-none d-lg-block.
Spacing and alignment	Bootstrap p-*, m-*, d-flex and text-* utilities combine with project-specific CSS.
Components	A collapsing navbar and an accordion provide navigation and answers. The bundle supplies their behavior.

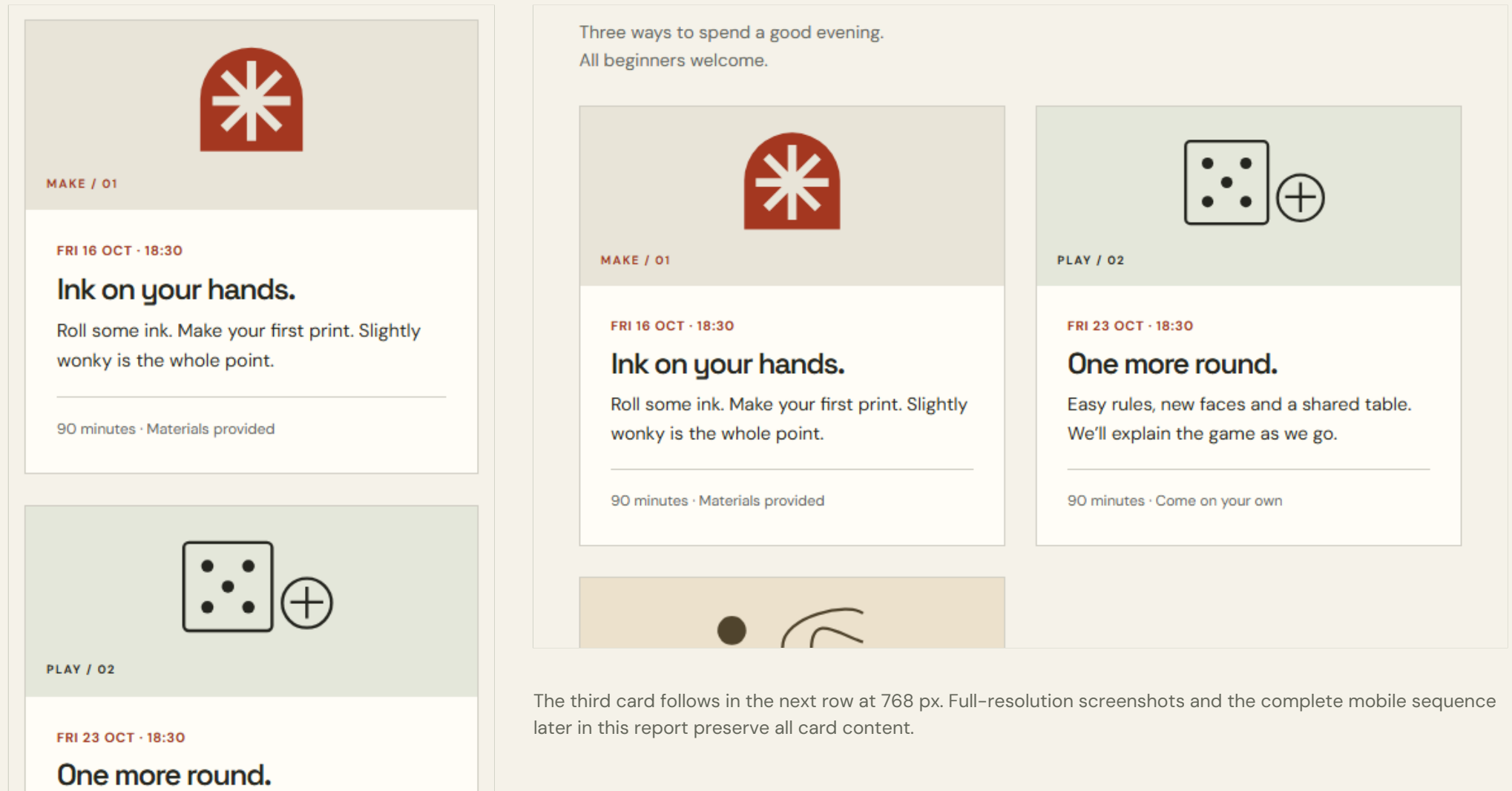
Grid classes and breakpoints

```
<div class="container">
  <div class="row g-4">
    <div class="col-12 col-md-6 col-lg-4">
      <article class="card">...</article>
    </div>
  </div>
</div>
```

Bootstrap lg starts at 992 px; the manual section changes to three columns at 64rem (1024 px with the default font size). The two thresholds are intentionally distinct. At the three required test widths, both card sections produce the required 1 / 2 / 3 arrangements.

Stacked cards become columns

Left: 375 px, a detail of the first two stacked cards. Right: 768 px, the first row of two cards. Crops retain the original proportions and arrangement.



The third card follows in the next row at 768 px. Full-resolution screenshots and the complete mobile sequence later in this report preserve all card content.

Three cards across at 1280 px

col-lg-4 gives each column four of Bootstrap's twelve grid units. The cards keep the same semantic article content.

02 / FIND YOUR PEOPLE

Next around the table.

Three ways to spend a good evening.
All beginners welcome.



MAKE / 01

FRI 16 OCT · 18:30

Ink on your hands.

Roll some ink. Make your first print. Slightly wonky is the whole point.

90 minutes · Materials provided



PLAY / 02

FRI 23 OCT · 18:30

One more round.

Easy rules, new faces and a shared table. We'll explain the game as we go.

90 minutes · Come on your own



NOTICE / 03

SAT 24 OCT · 11:00

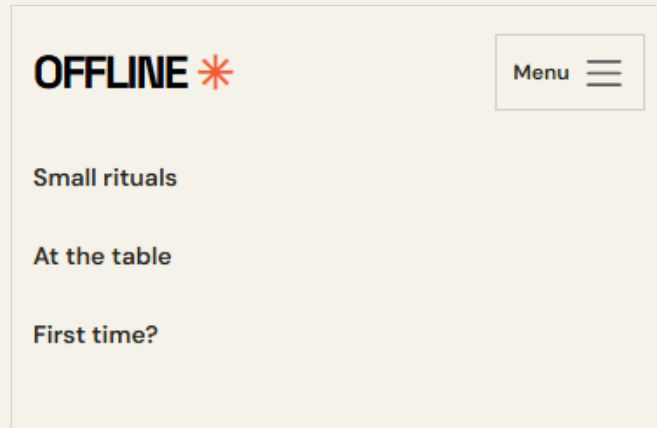
The slower route.

A small walk with no step target. Collect a few details you usually pass by.

45 minutes · Dress for the weather

Small-screen interactions

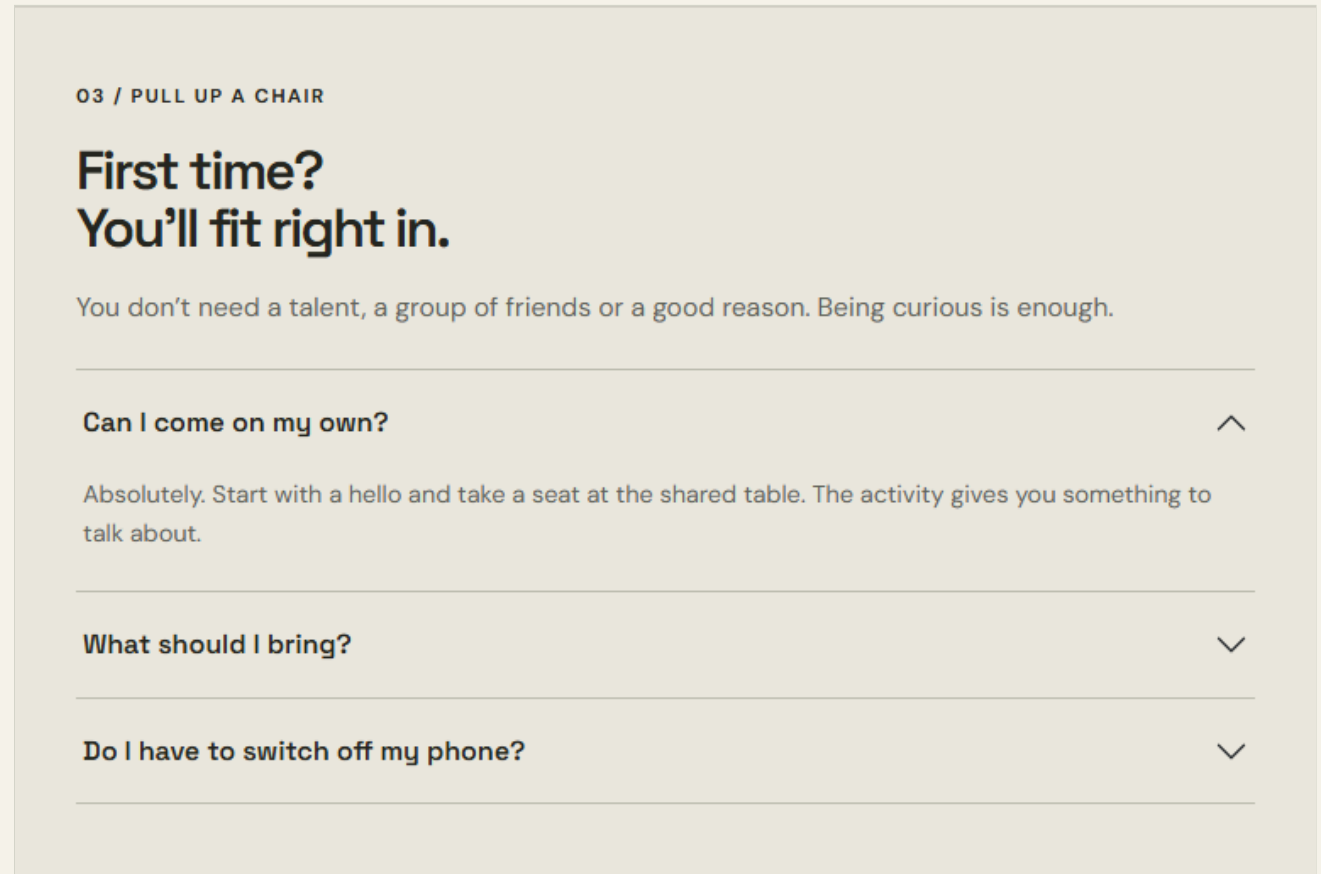
The navbar button opens the section links. The accordion expands answers. The hero keeps its important copy first when the columns stack.



Hero source order:

1. Introduction and actions
2. Poster illustration

At lg, order-lg-2 moves the copy to the right and order-lg-1 moves the poster left. On phones, the useful introduction stays first.



Bootstrap's bundle controls collapse state and aria-expanded. Native links navigate to section IDs. Custom CSS supplies the paper color, ink borders and visible keyboard focus.

Two ways to solve the same grid

Both sections place repeated cards in 1, 2 and 3 columns. The difference is where the responsive decisions live: the manual section states them in CSS, while the Bootstrap section selects an existing system through HTML classes.

Aspect	Task 1: custom CSS	Task 2: Bootstrap
Layout code	The ritual grid defines display: grid, its base track and two min-width overrides.	The gathering columns use col-12 col-md-6 col-lg-4 inside row and container.
Design control	48rem and 64rem are chosen for the ritual cards; spacing and heading changes share a query.	md and lg use Bootstrap's existing 768 px and 992 px thresholds; own CSS adds the visual style.
Change process	Changing the desktop track count means editing one grid-template-columns declaration.	Changing col-lg-4 to col-lg-6 gives two equal columns at lg; the class must change on each column.
Reuse and behavior	The grid needs no layout library. Any interactive behavior is a separate responsibility.	The same framework also supplies the navbar and accordion, with its JavaScript bundle.

When to choose each approach

Custom CSS fits a section that needs its own proportions, spacing changes or content-specific breakpoints, like the ritual grid. Bootstrap fits a page with repeated conventional columns and standard interactions, like the gathering cards, navigation and FAQ. Combining them works well here because the custom section has a clear boundary and the rest shares one responsive system.

The comparison refers to this submitted page. Neither approach automatically supplies the site's visual identity; the palette, typography, hierarchy and content still need design decisions.

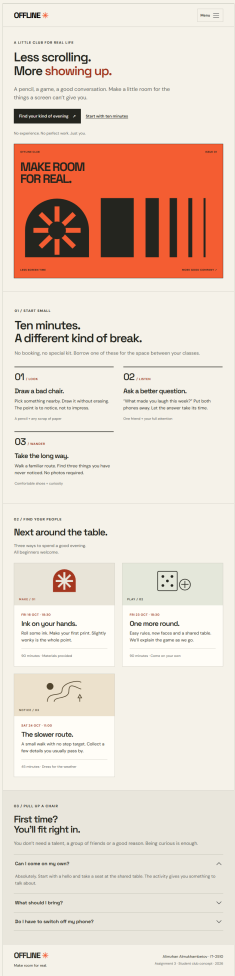
The complete page at three widths

375 px / 768 px / 1280 px. These full-page overviews show the entire document; the following spread makes the mobile page readable from top to bottom.

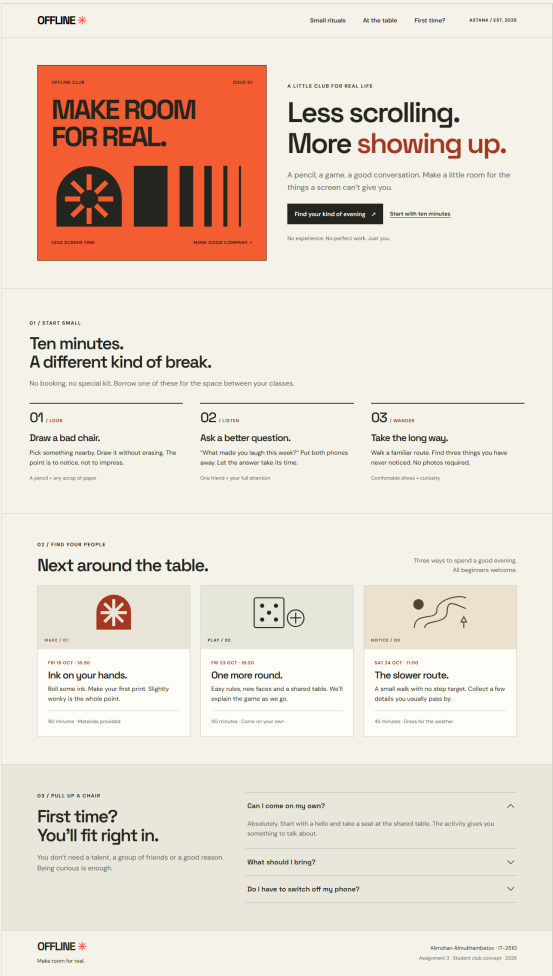
375 px
375 x 3876 px capture



768 px
768 x 3254 px capture



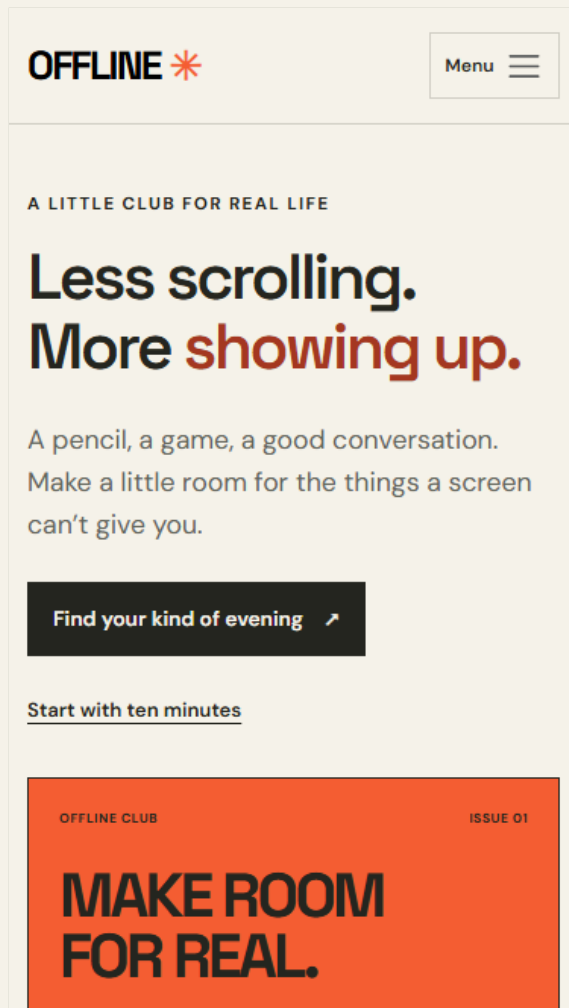
1280 px
1280 x 2264 px capture



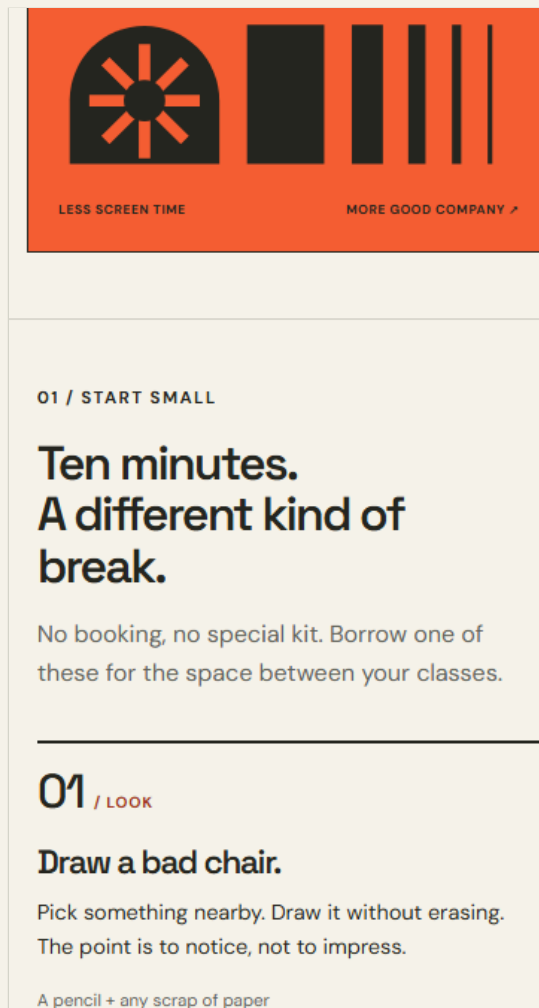
375 px: reading sequence 1 of 2

Read left to right, then continue on the next page. These are consecutive, non-overlapping crops of the same full-page screenshot; no content is rearranged or omitted.

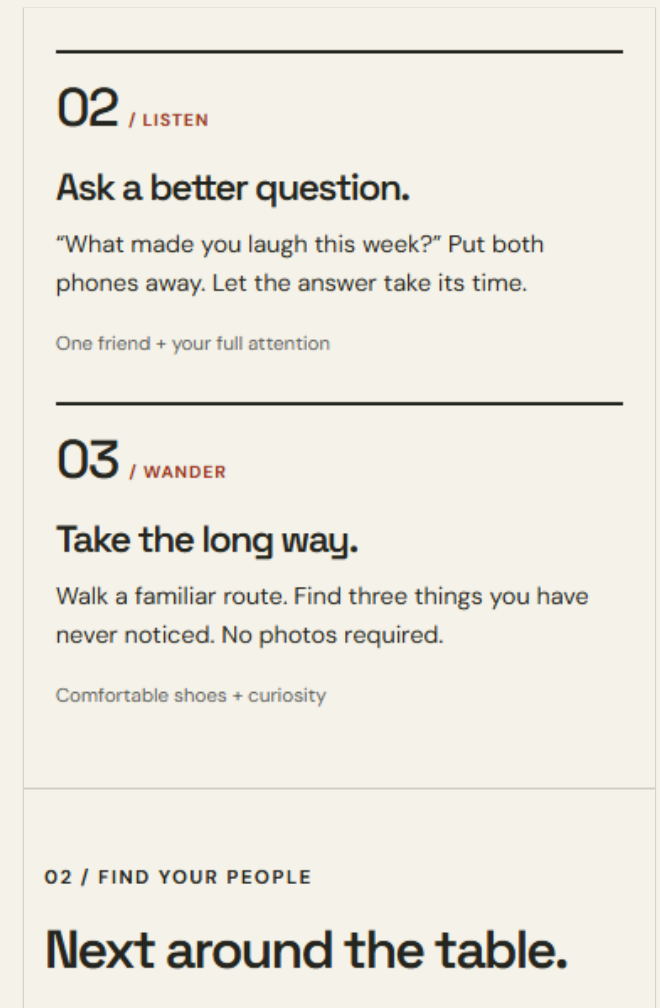
Part 1 of 6 / y 0–665 px



Part 2 of 6 / y 665–1340 px



Part 3 of 6 / y 1340–1938 px



375 px: reading sequence 2 of 2

Read left to right to reach the footer. These are consecutive, non-overlapping crops of the same full-page screenshot; no content is rearranged or omitted.

Part 4 of 6 / y 1938–2565 px

Three ways to spend a good evening.
All beginners welcome.



MAKE / 01

FRI 16 OCT · 18:30

Ink on your hands.

Roll some ink. Make your first print. Slightly wonky is the whole point.

90 minutes · Materials provided



PLAY / 02

FRI 23 OCT · 18:30

Part 5 of 6 / y 2565–3210 px

One more round.

Easy rules, new faces and a shared table.
We'll explain the game as we go.

90 minutes · Come on your own



NOTICE / 03

SAT 24 OCT · 11:00

The slower route.

A small walk with no step target. Collect a few details you usually pass by.

45 minutes · Dress for the weather

03 / PULL UP A CHAIR

Part 6 of 6 / y 3210–3876 px

First time? You'll fit right in.

You don't need a talent, a group of friends or a good reason. Being curious is enough.

Can I come on my own? ^

Absolutely. Start with a hello and take a seat at the shared table. The activity gives you something to talk about.

What should I bring? v

Do I have to switch off my phone? v

OFFLINE *

Make room for real.

Alimzhan Almukhambetov · IT-2510
Assignment 3 · Student club concept · 2026

What the implementation demonstrates

The page combines a handmade CSS grid with Bootstrap's grid and components while keeping one visual style. The main takeaway is that responsive design changes the arrangement and priority of content, not only its size: columns stack, the hero preserves reading order, and secondary content can disappear on small screens.

Another takeaway is the importance of the cascade. Bootstrap is loaded first, the project stylesheet follows it, and a mobile-first query adds changes as the viewport grows. Testing the actual page at 375, 768 and 1280 px makes these decisions visible and helps catch horizontal overflow or an inaccessible menu.

Verification

Check	Observed result: passed
375 / 768 / 1280 px	Manual and Bootstrap card grids show 1 / 2 / 3 columns; no horizontal page scroll.
Navbar and accordion	Menu toggles on a phone; navigation targets exist; answers expand and collapse.
Source and dependencies	Viewport meta tag, official Bootstrap CDN links, local CSS after Bootstrap, bundle before <code></body></code> .

The hosted page passed all 97 recorded checks at 8 viewport widths, including the required sizes and breakpoint boundaries. The browser evidence was captured on 3 October 2026; full results are in [evidence/deployed/verification.json](#).

Credits and references

The topic and visual language reuse the existing OFFLINE midterm project with permission. Space Grotesk and DM Sans are stored locally under their included SIL Open Font License files. Bootstrap is used under its MIT license. AI tools assisted with implementation, report drafting and verification.

Bootstrap 5.3: introduction and CDN setup

Bootstrap 5.3: grid system

Bootstrap 5.3: navbar

Bootstrap 5.3: accordion

W3Schools: CSS media queries

Course material: G. A. Abitova, Web Technologies Front-End Development, Part 2 (2022); Assignment 3 brief supplied for WEB Technologies 1.